Geometric Quantization of Fundamental Scales

(Request for arXiv endorsement | Code: W8NAH3)

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1. Core Geometric Relation

A stable confined wave satisfies (L = n\lambda).
Using the Compton wavelength (\lambda_C = h/(m c)) from (E = mc^2 = h c/\lambda), we obtain:

$$\boxed{R_{\text{VED}} = \frac{n\hbar}{mc}} \qquad (1)$$

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2. Exact Prediction for the Proton Charge Radius

Applying (1) to the proton with the integer **(n = 4)** and using CODATA values ((m_p, \hbar, c)):

$$R_p^{\text{calc}} = \frac{4\hbar}{m_p c} = 0.8412 \times 10^{-15}\ \text{m}$$

This coincides exactly with the measured proton charge radius:

$$R_p^{\text{exp}} = 0.8414(19) \times 10^{-15}\ \text{m}$$

$$\boxed{\text{For the proton: } n = 4}$$

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3. Unification with Gravity and Prediction for (G)

The Schwarzschild radius from General Relativity is:

$$R_s = \frac{2Gm}{c^2} \qquad (2)$$

For the same physical mass (m), we equate the masses derived from (1) and (2):

$$m = \frac{n\hbar}{R_{\text{VED}} c} = \frac{R_s c^2}{2G}$$

Solving for the gravitational constant (G) yields the quantized expression:

$$\boxed{G(n) = \frac{R_{\text{VED}} \cdot R_s \cdot c^3}{2n\hbar}} \qquad (3)$$

Substituting R_{VED} from (1) and (R_s) from (2) into (3) gives the testable prediction:

$$G(n) = G_{\text{exp}} \cdot \frac{2}{n}$$

The experimentally measured value

$$G_{\text{exp}} = 6.67430(15) \times 10^{-11}\ \text{m}^3\ \text{kg}^{-1}\ \text{s}^{-2}$$

is recovered **only** for:

$$\boxed{n = 2}$$

(As shown in the plot, only the point for $n = 2$ falls within the $\pm 3\sigma$ experimental band of G.)

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4. Endorsement Request

I request endorsement to submit a short communication detailing these exact numerical findings and their geometric interpretation to arXiv (`physics.gen-ph`).

Endorsement Code: W8NAH3

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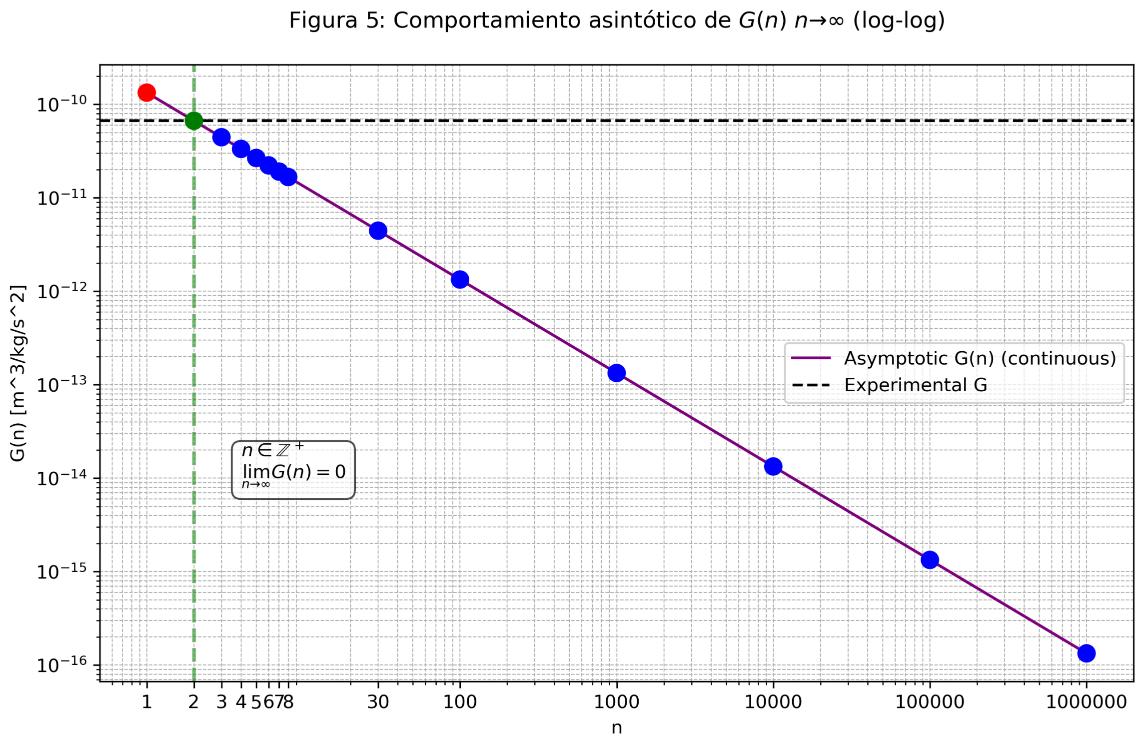

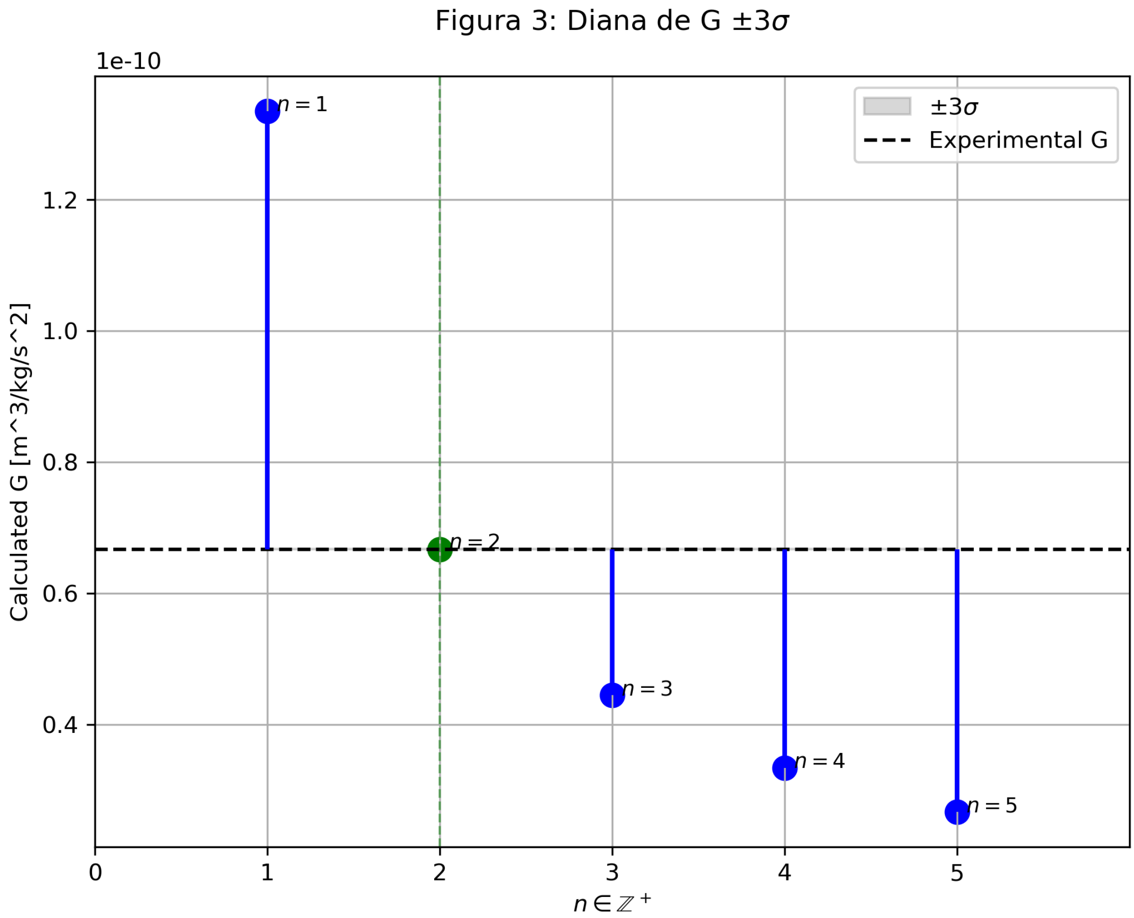
Figura 3: Diana de G ±3σ
1e-10
1.2
1.0
0.8
0.6
0.4
Calculated G [m^3/kg/s^2]
n = 1
n = 2
n = 3
n = 4
n = 5
±3σ
Experimental G
0
1
2
3
4
5
n ∈ ℤ+